Normalization Parameters

Instead of restricting the Activations of each layer to be *strictly* **Zero** mean and **unit** variance, Normalization allows the network to **learn parameters** γ and **β** that can convert the mean and variance to any value that **Minimizes Loss.**

Input : Values of x over a mini – batch : $\mathcal{B} = \{x_{1\ldots m}\}$;
Parameters to be learned : $\gamma,\ \beta$

Output : $\{y_i = \mathrm{BN}_{\gamma,\beta}(x_i)\}$

$$\mu_{\mathcal{B}} \leftarrow \frac{1}{m}\sum_{i=1}^{m} x_i \qquad \text{// mini-batch mean}$$

$$\sigma_{\mathcal{B}}^2 \leftarrow \frac{1}{m}\sum_{i=1}^{m}(x_i - \mu_{\mathcal{B}})^2 \qquad \text{// mini-batch variance}$$

$$\widehat{x}_i \leftarrow \frac{x_i - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}} \qquad \text{// normalize}$$

$$y_i \leftarrow \gamma \widehat{x}_i + \beta \equiv \mathrm{BN}_{\gamma,\beta}(x_i) \qquad \text{// scale and shift}$$